Quick Contact Detection for Simplified Rod-Cone Docking Process

Mengqi Yang, Beijing Institute of Technology

2026/9/4

1. Introduction

We have already discussed a simplified kinematic model for rod-cone contact process in the blog “A Simplified Kinematic Strategy for Rod-Cone Docking Contact”. In the strategy, we transform the rod-cone contact to virtual-rigid contact model for flexible truss described using Floating Frame of Reference Formulation (FFRF). Such virtual rigid contact also requires contact detection. To avoid complex detection algorithm during code developing like AABB or signal distance field (SDF), we propose a simplified strategy in this document.

2. Review on the Kinematic Description

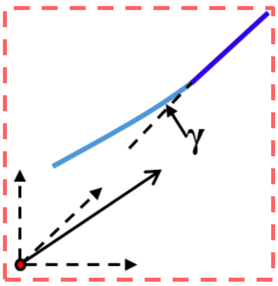

Fig. 1. Virtual rigid contact model: Shallow-blue curve stands for a flexible body, while the deep-blue one stands for the virtual rigid body

Still, we use space $\mathbb{Q}=\mathbb{R}^3\times\mathrm{SO}(3)$ to model the rigid body motion. The displacement $\mathbf{r}$ of one point on the flexible body reads

$$\mathbf{r}=\mathbf{p}+\mathbf{R}\left(\bar{\mathbf{x}}_0+\bar{\mathbf{u}}_f\left(\bar{\mathbf{x}}_0\right)\right) \tag{1}$$

where $\mathbf{p}$ denotes the displacement of the reference point on the flexible body, $\mathbf{R}$ is rotation matrix of the reference point, $\bar{\mathbf{x}}_0$ is the static local position of points on the flexible body referred to the reference point, and $\bar{\mathbf{u}}_f(\bar{\mathbf{x}}_0)$ is the flexible deformation at point $\bar{\mathbf{x}}_0$ described in the reference frame $\mathbf{R}$.

As shown in Fig. 1, we assume that the flexible bodies are attached by massless rigid parts. And when contact happens, the forces are only computed on the rigid parts. In this way, we avoid the refined model of rigid body. That's why we call this method a "virtual" rigid contact model.

Without the loss of generality, we take a simplified rod-cone geometry model in Fig. 2 for example. In this model, the contact can be formulated by only considering point-point contact when relative bias angle is small enough.

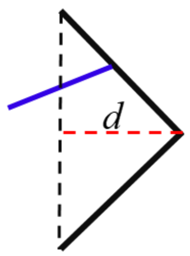

Fig. 2. Simplified rod-cone geomerty model: Blue line stands for the virtural rigid rod and black cone stands for the virtual rigid cone

Based on Eq. (1), we further derive the position of rod tip,

$$\mathbf{r}_t = \mathbf{p}_t + \mathbf{R}\left(\bar{\mathbf{x}}_t + \bar{\mathbf{u}}_f\left(\bar{\mathbf{x}}_t\right)\right) \tag{2}$$

Still, $\bar{\mathbf{x}}_t$ stands for the static position in the reference coordinates. And it should be noticed that $\bar{\mathbf{u}}_f(\bar{\mathbf{x}}_t)$ is caused only by the flexible parts. By using the beam model in Fig. 1, $\bar{\mathbf{u}}_f(\bar{\mathbf{x}}_t)$ can be obtained. We use $\boldsymbol{\gamma} = \begin{bmatrix}\gamma_x & \gamma_y & \gamma_z\end{bmatrix}^{\mathrm{T}}$ to represent the small deformation angles of beam. γ_x is the torsion angle while γ_y and γ_z are deflection angles in two directions respectively. If we make the contact model simpler, the rod is limited to a line, then the torsion effect can be ignored. Such that we have

$$\bar{\mathbf{u}}_f\left(\bar{\mathbf{x}}_t\right)=\bar{\mathbf{u}}_f\left(\bar{\mathbf{x}}_n\right)+\begin{bmatrix}0\cdot\gamma_x\\ d\cdot\gamma_y\\ d\cdot\gamma_z\end{bmatrix}=\bar{\mathbf{u}}_f\left(\bar{\mathbf{x}}_n\right)+\mathrm{diag}\left(0,d,d\right)\boldsymbol{\gamma}_n=\bar{\mathbf{u}}_f\left(\bar{\mathbf{x}}_n\right)+\mathbf{D}\boldsymbol{\gamma}_n \tag{3}$$

where d is the length of rod and $\bar{\mathbf{x}}_n$ is the position of nearest beam node. If we consider a finite element model,

$$\boldsymbol{\gamma}_n=\mathbf{S}_\gamma\left(\bar{\mathbf{x}}_n\right)\mathbf{e}_n \tag{4}$$

Then Eq. (3) can be derived as

$$\bar{\mathbf{u}}_f\left(\bar{\mathbf{x}}_t\right)=\bar{\mathbf{u}}_f\left(\bar{\mathbf{x}}_n\right)+\mathbf{D}\mathbf{S}_\gamma\left(\bar{\mathbf{x}}_n\right)\mathbf{e}_n=\left[\mathbf{S}\left(\bar{\mathbf{x}}_n\right)+\mathbf{D}\mathbf{S}_\gamma\left(\bar{\mathbf{x}}_n\right)\right]\mathbf{e}_n=\mathbf{S}_r\left(\bar{\mathbf{x}}_n\right)\mathbf{e}_n \tag{5}$$

The direction of rod can also be quickly determined by make difference on the two-ends of corresponding nodes. We can then obtain the velocity of rod tip

$$\begin{aligned}\dot{\mathbf{r}}_t&=\dot{\mathbf{p}}+\dot{\mathbf{R}}\left(\bar{\mathbf{x}}_t+\bar{\mathbf{u}}_f\right)+\mathbf{R}\dot{\bar{\mathbf{u}}}_f\\&=\begin{bmatrix}\mathbf{I} & -\mathbf{R}\left(\bar{\mathbf{x}}_t+\mathbf{S}_r\left(\bar{\mathbf{x}}_n\right)\mathbf{e}_n\right)^\wedge & \mathbf{R}\mathbf{S}_r\left(\bar{\mathbf{x}}_n\right)\end{bmatrix}\begin{bmatrix}\dot{\mathbf{p}}\\ \bar{\boldsymbol{\omega}}\\ \dot{\mathbf{e}}_n\end{bmatrix}\end{aligned} \tag{6}$$

Follow the same process, the position of arbitrary point on the cone can also be formulated. A slight difference lies in the position of cone that considers the contribution of torsion, which only requires the modification of matrix $\mathbf{D}$.

$$\bar{\mathbf{u}}_f\left(\bar{\mathbf{x}}_c\right)=\bar{\mathbf{u}}_f\left(\bar{\mathbf{x}}_n\right)+\begin{bmatrix}0\cdot\gamma_x\\ d\cdot\gamma_y-t\left(\bar{\mathbf{x}}_n\right)\cdot i_z\left(\bar{\mathbf{x}}_n\right)\gamma_x\\ d\cdot\gamma_z+t\left(\bar{\mathbf{x}}_n\right)\cdot i_y\left(\bar{\mathbf{x}}_n\right)\gamma_x\end{bmatrix} \tag{7}$$

where $t\left(\bar{\mathbf{x}}_n\right)$ is the width of cone at point $\bar{\mathbf{x}}_n$, $i_y\left(\bar{\mathbf{x}}_n\right)$ and $i_z\left(\bar{\mathbf{x}}_n\right)$ respectively stand for the local direction in cone at point $\bar{\mathbf{x}}_n$. Additionally, we denote $\mathbf{d}_1$ as the direction of centerline for the cone.

3. Contact Detection Algorithm

Actually, the rod-cone contact problem has between transformed to a point-surface contact problem with regular geometry, which allows us use analytical geometric method to achieve low-cost code development. Here we introduce the contact detection algorithm with two steps. The two-step algorithm follows the same logic with popular contact detection in industry implementation.

3.1.Rough Scan

Similar with the AABB method, we can also use analytical method for rough scan process. We firstly discuss ideal cases, where rod tip has already been put inside the cone as shown in Fig. 3. To find whether the rod and cone has contacted, we shall firstly find the displacement of rod tip expressed using the local frame located at the cone:

$$\bar{\boldsymbol{\chi}} = \mathbf{R}_c^{\mathrm{T}} \left(\mathbf{r}_t - \mathbf{r}_c \right) \tag{8}$$

Notice that the attitude of cone is influenced by the deformation of truss, that

$$\mathbf{R}_{\text{cone}}^{\text{rigid}} = \mathbf{R}_{\text{cone}}^{\text{rigid}} \exp_{\text{SO(3)}} \left(\boldsymbol{\gamma}_c \right) \tag{9}$$

To better describe the geometry, we let $\chi_1 = \bar{\boldsymbol{\chi}} \cdot \mathbf{d}_1$ as the projection of Eq. (8) to the centerline of the cone. And then we have its corresponding radius $t = t\left(\bar{\chi}_1\right)$. With such geometry parameter, the rough scan can be simplified using

$$\tau = \left\| \bar{\boldsymbol{\chi}} - \bar{\chi}_1 \mathbf{d}_1 \right\| < t\left(\bar{\chi}_1\right)? \tag{10}$$

where τ stand for the projected rod tip on plane determined by $\bar{\chi}_1$, which is vertical to the centerline. And if $\tau < t$, the rod tip has not contacted with the cone, otherwise we move on to next step to find the contact point. And if $\bar{\chi}_1 > d$, the rod has not entered the cone.

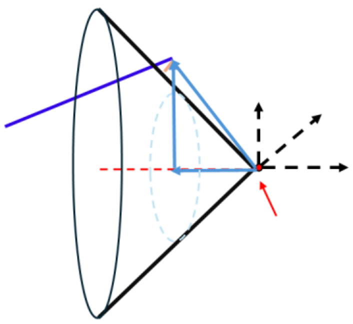

Fig. 3. Ideal cases with rod inside the cone

By the way, if the cases were not very ideal, for arbitrary situation, we can only detect the tip and the bottom of the rod, to find whether contact has happened, using the same rough scan process, very similar with AABB.

3.2. Refined Detection

To determine which point on the cone is the contact point, we have to find the minimal distance for the contact penetration. As previously discussed, the local displacement of rod tip can be divided into centerline entry and the plane entries. We can find the corresponding direction of contact point compared with the center of cone projected on the plane. It is worth noting that the direction of rod is not very important as here we only consider a point-surface contact problem. Then we define

$$\boldsymbol{\theta} = \frac{\bar{\boldsymbol{\chi}} - \bar{\chi}_1 \mathbf{d}_1}{\tau} = \begin{bmatrix} 0 \\ \theta_1 \\ \theta_2 \end{bmatrix} \qquad (11)$$

$$p = \tau - t\left(\bar{\chi}_1\right)$$

where $\boldsymbol{\theta}$ is the local direction, and p is the penetration depth.

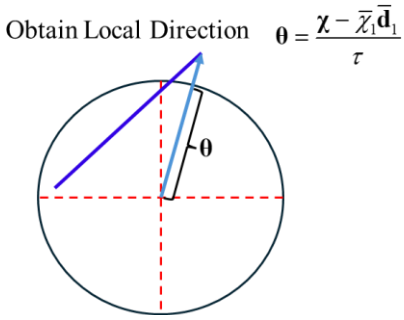

Fig. 4. Contact direction compared with the centerline

However, the point with corresponding direction on the circle $\bar{\chi}_1$ is not the closest point of contact for the three-dimensional geometry. We shall project the contact point back to three-dimensional geometry show in Fig. 3. Let half of the cone angle to be υ, and we let $s = \tan \upsilon$ so that the projected contact point can be found using

$$\bar{\mathbf{x}}^{\mathrm{c}}_{\mathrm{cone}} = \left(\bar{\chi}_1 + \frac{ps}{1+s^2}\right)\mathbf{d}_1 + s\left(\bar{\chi}_1 + \frac{ps}{1+s^2}\right)\boldsymbol{\theta} \qquad (12)$$

The penetration depth is found to be $\delta = \frac{p}{\sqrt{1+s^2}}$. Also, it is important to find the velocity of the contact point on the cone to find the friction force. With the expression using Eq. (12), the previously discussed Eq. (7) is not very useful. We have to construct a suitable expression using Eq. (11) and Eq. (12).

We use the following expression to fine the global position of contact point on the cone:

$$\mathbf{r}_{\text{cone}}^{\text{c}} = \mathbf{p}_{\text{cone}}^{\text{rigid}} + \mathbf{R}_{\text{cone}}^{\text{rigid}} \left(\overline{\mathbf{x}}_{\text{cone}}^{\text{rigid}} + \mathbf{u}_f \left(\overline{\mathbf{x}}_{\text{cone}}^{\text{rigid}} \right) \right) + \mathbf{R}_{\text{cone}} \overline{\mathbf{x}}_{\text{cone}}^{\text{c}} \tag{13}$$

So that the velocity can be expressed using

$$\begin{aligned} \dot{\mathbf{r}}_{\text{cone}}^{\text{c}} &= \dot{\mathbf{p}}_{\text{cone}}^{\text{rigid}} + \dot{\mathbf{R}}_{\text{cone}}^{\text{rigid}} \left(\overline{\mathbf{x}}_{\text{cone}}^{\text{rigid}} + \overline{\mathbf{u}}_f \right) + \mathbf{R} \dot{\overline{\mathbf{u}}}_f + \dot{\mathbf{R}}_{\text{cone}}^{\text{c}} \overline{\mathbf{x}}_{\text{cone}}^{\text{c}} \\ &= \begin{bmatrix} \mathbf{I} & -\mathbf{R}_{\text{cone}}^{\text{rigid}} \left(\overline{\mathbf{x}}_{\text{cone}}^{\text{rigid}} + \mathbf{S}_r \left(\overline{\mathbf{x}}_n \right) \mathbf{e}_n + \exp_{\text{SO(3)}} \left(\boldsymbol{\gamma}_c \right) \overline{\mathbf{x}}_{\text{cone}}^{\text{c}} \right)^{\wedge} & \mathbf{R} \mathbf{S}_r \left(\overline{\mathbf{x}}_n \right) \end{bmatrix} \begin{bmatrix} \dot{\mathbf{p}}_{\text{cone}}^{\text{rigid}} \\ \overline{\boldsymbol{\omega}}_{\text{cone}}^{\text{rigid}} \\ \dot{\mathbf{e}}_n \end{bmatrix} \\ &\quad + \mathbf{R}_{\text{cone}} \mathbf{T}_{\text{SO(3)}} \left(\boldsymbol{\gamma}_c \right) \dot{\boldsymbol{\gamma}}_c \overline{\mathbf{x}}_{\text{cone}}^{\text{c}} \end{aligned} \tag{14}$$

To conclude, the analytical for rod-cone contact follows this work flow:

1. Determine the geometry of cone, especially keep the parameter $s = \tan \upsilon$ and cone height d;
2. Find global position of rod tip using Eq. (3);
3. Conduct the rough scan step using Eq. (8)-Eq. (10), and check if $\overline{\chi}_1 > d$. If contacted, go to step 4, otherwise stop;
4. Find contact point on the cone using Eq. (11) and Eq. (12).
5. Obtain global position and velocity of contact point using Eq. (13) - Eq. (14).
6. Find global contact direction and relative velocity;
7. Compute contact force with friction.